Exercise Sheet 8 - Solutions
Predicate Logic – Natural Deduction

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1. $\forall x.\forall y.\mathtt{max}(x,y) \geq \mathtt{min}(x,y)$
2. $\forall x.\neg\exists y.\neg(x = y) \land \mathtt{max}(x,y) = \mathtt{min}(x,y)$
3. $$\dfrac{\dfrac{\dfrac{\overline{\exists x.p(x)}^{\,2} \qquad \dfrac{\dfrac{\dfrac{\overline{\forall x.p(x) \to q(x)}^{\,1}}{p(x) \to q(x)}\,[\forall E] \qquad \overline{p(x)}^{\,3}}{q(x)}\,[\to E]}{\exists x.q(x)}\,[\exists I]}{\exists x.q(x)}\,3\;[\exists E]}{(\exists x.p(x)) \to \exists x.q(x)}\,2\;[\to I]}{(\forall x.p(x) \to q(x)) \to (\exists x.p(x)) \to \exists x.q(x)}\,1\;[\to I]$$
4. $$\dfrac{\dfrac{\dfrac{\overline{\exists x.\exists y.p(x) \land r(x,y)}^{\,2} \qquad \dfrac{\overline{\exists y.p(x) \land r(x,y)}^{\,3} \qquad \dfrac{\dfrac{\dfrac{\overline{p(x) \land r(x,y)}^{\,4}}{r(x,y)}\,[\land E_R]}{\exists y.r(x,y)}\,[\exists I] \qquad \dfrac{\dfrac{\overline{\forall x.p(x) \to \neg\exists y.r(x,y)}^{\,1}}{p(x) \to \neg\exists y.r(x,y)}\,[\forall E] \qquad \dfrac{\overline{p(x) \land r(x,y)}^{\,4}}{p(x)}\,[\land E_L]}{\neg\exists y.r(x,y)}\,[\to E]}{\bot}\,[\neg E]}{\bot}\,4\;[\exists E]}{\bot}\,3\;[\exists E]}{\neg\exists x.\exists y.p(x) \land r(x,y)}\,2\;[\neg I]}{(\forall x.p(x) \to \neg\exists y.r(x,y)) \to \neg\exists x.\exists y.p(x) \land r(x,y)}\,1\;[\to I]$$
5. $$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\forall x.\forall y.x > y \lor \neg(x > y)}{\forall y.x + z > y \lor \neg(x + z > y)}\,[\forall E]}{x + z > y + z \lor \neg(x + z > y + z)}\,[\forall E] \qquad \Pi_1 \quad \Pi_3}{\mathtt{min}(x + z, y + z) \geq z}\,[\lor E]}{\forall z.\mathtt{min}(x + z, y + z) \geq z}\,[\forall I]}{\forall y.\forall z.\mathtt{min}(x + z, y + z) \geq z}\,[\forall I]}{\forall x.\forall y.\forall z.\mathtt{min}(x + z, y + z) \geq z}\,[\forall I]$$

   where Π_1 is:

   $$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\forall x.\forall y.\forall z.x = y \to y \geq z \to x \geq z}{\forall v.\forall w.\mathtt{min}(x + z, y + z) = v \to v \geq w \to \mathtt{min}(x + z, y + z) \geq w}\,[\forall E]}{\forall w.\mathtt{min}(x + z, y + z) = y + z \to y + z \geq w \to \mathtt{min}(x + z, y + z) \geq w}\,[\forall E]}{\mathtt{min}(x + z, y + z) = y + z \to y + z \geq z \to \mathtt{min}(x + z, y + z) \geq z}\,[\forall E] \qquad \Pi_2}{y + z \geq z \to \mathtt{min}(x + z, y + z) \geq z}\,[\to E] \qquad \dfrac{\dfrac{\forall x.\forall y.x + y \geq y}{\forall w.y + w \geq w}\,[\forall E]}{y + z \geq z}\,[\forall E]}{\mathtt{min}(x + z, y + z) \geq z}\,[\to E]}{x + z > y + z \to \mathtt{min}(x + z, y + z) \geq z}\,1\;[\to I]$$

where Π_2 is

$$\dfrac{\dfrac{\dfrac{\forall x.\forall y.x > y \to \texttt{min}(x, y) = y}{\forall y.x + z > y \to \texttt{min}(x + z, y) = y}\;[\forall E]}{x + z > y + z \to \texttt{min}(x + z, y + z) = y + z}\;[\forall E] \qquad \dfrac{}{x + z > y + z}\;1}{\texttt{min}(x + z, y + z) = y + z}\;[\to E]$$

where Π_3 is:

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\forall x.\forall y.\forall z.x = y \to y \geq z \to x \geq z}{\forall v.\forall w.\texttt{min}(x + z, y + z) = v \to v \geq w \to \texttt{min}(x + z, y + z) \geq w}\;[\forall E]}{\forall w.\texttt{min}(x + z, y + z) = x + z \to x + z \geq w \to \texttt{min}(x + z, y + z) \geq w}\;[\forall E]}{\texttt{min}(x + z, y + z) = x + z \to x + z \geq z \to \texttt{min}(x + z, y + z) \geq z}\;[\forall E] \qquad \Pi_4}{x + z \geq z \to \texttt{min}(x + z, y + z) \geq z}\;[\to E] \qquad \dfrac{\dfrac{\forall x.\forall y.x + y \geq y}{\forall y.x + y \geq y}\;[\forall E]}{x + z \geq z}\;[\forall E]}{\texttt{min}(x + z, y + z) \geq z}\;[\to E]}{\neg(x + z > y + z) \to \texttt{min}(x + z, y + z) \geq z}\;2\ [\to I]$$

where Π_4 is

$$\dfrac{\dfrac{\dfrac{\forall x.\forall y.\neg(x > y) \to \texttt{min}(x, y) = x}{\forall y.\neg(x + z > y) \to \texttt{min}(x + z, y) = x + z}\;[\forall E]}{\neg(x + z > y + z) \to \texttt{min}(x + z, y + z) = x + z}\;[\forall E] \qquad \dfrac{}{\neg(x + z > y + z)}\;2}{\texttt{min}(x + z, y + z) = x + z}\;[\to E]$$